StockWise AI

Intelligent Inventory Optimization & Demand Intelligence

Comprehensive Operational & Executive Inventory Audit Report

Generated: October 29, 2026

REPORT SCOPE & EXECUTIVE SPECIFICATIONS

- Total Active SKU Portfolio: 200 items across 6 retail categories
- Total Inventory Capital at Cost: $805,868.5
- Verified Suppliers Analyzed: 24 global and regional vendor partners
- Stockout Exposure Detected: 127 products (31 critical urgency)
- Excess Capital Locked: $153,930.8 in overstocked inventory

EXECUTIVE PREFACE

This intelligence report provides an algorithmic analysis of inventory positions, empirical demand velocity, lead-time variability, and service-level trade-offs across the catalog. By replacing static reorder rules with dynamic safety stock formulas and demand forecasting, StockWise AI isolates actionable opportunities to mitigate stockouts while systematically shedding excess non-moving capital.

DEMONSTRATION & SYNTHETIC DATA NOTICE

All product records, sales figures, and supplier metrics shown herein represent synthetic test data for system demonstration purposes. Figures should not be construed as actual retail financial records.

1. Executive Inventory Performance Summary

StockWise AI analyzed 200 active products representing $805,868.5 in invested inventory cost ($1,805,231.46 retail valuation). The analysis isolates a structural bi-modal inefficiency: while 127 products (63.5%) are facing imminent stockout risk resulting in an estimated $216,815.62 in lost sales exposure, another 11 products hold $153,930.8 in unproductive overstocked inventory. Current average inventory cover stands at 24.1 days.

Portfolio Key Performance Indicators

TOTAL PRODUCTS
200
Active Catalog SKUs

STOCKOUT RISK
127
31 Critical Urgency

OVERSTOCKED ITEMS
11
Exceeding 75d cover

SLOW-MOVING
0
Low daily velocity

AVG INVENTORY COVER
24.1 d
Days of available stock

INVENTORY COST VALUE
$805.9k
$1805.2k Retail

SUPPLIER ON-TIME
90.7%
Weighted vendor reliability

LOST SALES EXPOSURE
$216.8k
Past 30d unfulfilled demand

Category Health & Exposure Matrix

Category	SKU Count	Stockout Risk	Risk Ratio	Excess Value	Strategic Status
Electronics	40	28	70.0%	$87,466	High Attention Required
Apparel & Footwear	35	24	68.6%	$38,227	High Attention Required
Home & Kitchen	35	17	48.6%	$8,546	High Attention Required
Health & Personal Care	30	18	60.0%	$2,866	High Attention Required
Grocery & Gourmet	30	17	56.7%	$525	High Attention Required
Sports & Outdoors	30	23	76.7%	$16,301	High Attention Required

2. Statistical Demand Forecasting Findings

The StockWise AI forecasting engine applies adaptive statistical models (Holt Linear Exponential Smoothing, Weighted Moving Averages, and 14-day Rolling Averages) over 12 months of empirical demand history. Forecast confidence intervals (90% to 99% Z-scores) integrate both customer demand volatility and vendor delivery lead-time dispersion. Holiday surges (Q4) and seasonal shifts are decomposed from baseline steady-state velocities.

3. Stockout Risk Detections & Immediate Vulnerabilities

Products flagged below have available inventory insufficient to cover supplier lead time demand (DOC < Lead Time). Immediate purchase order generation is

Product Name	Category	Avail Stock	Daily Demand	Cover	Lead Time	Risk	Reorder Qty	PO Cost
Quantum Sound Wireless Noi...	Electronics	1 pcs	11.45 /d	0.1 d	18 d	Critical	610 units	$41,480
Ionic High-Speed Hair Drye...	Health & Personal Care	1 pcs	10.88 /d	0.1 d	20 d	Critical	595 units	$38,675
AutoClean Robot Vacuum LiD...	Electronics	2 pcs	11.67 /d	0.2 d	28 d	Critical	790 units	$173,800
Water-Resistant Trail Runn...	Apparel & Footwear	2 pcs	11.03 /d	0.2 d	16 d	Critical	520 units	$27,040
Damascus Steel 8" Chef Kni...	Home & Kitchen	2 pcs	10.73 /d	0.2 d	35 d	Critical	830 units	$34,860
Cold-Extracted Extra Virgi...	Grocery & Gourmet	2 pcs	11.52 /d	0.2 d	16 d	Critical	540 units	$7,560
All-Season Ultralight 2-Pe...	Sports & Outdoors	2 pcs	11.22 /d	0.2 d	28 d	Critical	765 units	$95,625
AeroRoad Lightweight Cycli...	Sports & Outdoors	2 pcs	11.42 /d	0.2 d	32 d	Critical	815 units	$47,270

4. Overstock & Slow-Moving Capital Optimization

Identified 11 products holding inventory cover exceeding 75 days. Over $153,930.8 in working capital is currently trapped in slow-moving stock, generating ur

Product Name	Category	Current Stock	Days Cover	Excess Units	Excess Cost	Recommended Action
ThunderDock 12-in-1 Triple...	Electronics	461 units	495.7 d	422 units	$32,916	Halt new orders. Initiate targeted p...
RetroStudio Over-Ear Studi...	Electronics	318 units	383.1 d	288 units	$27,360	Halt new orders. Initiate targeted p...
Classic Trench Coat Double...	Apparel & Footwear	310 units	356.3 d	277 units	$18,836	Halt new orders. Initiate targeted p...
HomeTheater Full HD Mini P...	Electronics	146 units	167.8 d	114 units	$12,540	Halt new orders. Initiate targeted p...
Full-Grain Artisan Leather...	Apparel & Footwear	180 units	225 d	147 units	$11,613	Halt new orders. Initiate targeted p...
Golf Rangefinder with Slop...	Sports & Outdoors	229 units	279.3 d	199 units	$10,348	Halt new orders. Initiate targeted p...

5. Supplier Lead-Time & Vendor Reliability Audit

Supplier Name	Origin	Specialty	Lead Time	Std Dev	On-Time	Rating	Vendor Tier
Apex Tech Electronics Ltd	Taiwan	Electronics	14 days	±3.2 d	94.0%	4.8 / 5.0	Tier 1 Preferred
Silicon Core Semiconduc...	South Korea	Electronics	28 days	±6.5 d	82.0%	3.9 / 5.0	Tier 3 Review
Nordic Audio Systems	Sweden	Electronics	18 days	±2.8 d	96.0%	4.9 / 5.0	Tier 1 Preferred
Shenzhen Micro Innovati...	China	Electronics	21 days	±5.1 d	86.0%	4.1 / 5.0	Tier 3 Review
Milanese Weavers & Mills	Italy	Apparel & Footwear	25 days	±4 d	91.0%	4.6 / 5.0	Tier 2 Standard
Bengal Cotton Crafts	India	Apparel & Footwear	19 days	±3.5 d	88.0%	4.2 / 5.0	Tier 2 Standard
Iberian Leather Guild	Spain	Apparel & Footwear	30 days	±5.8 d	84.0%	4 / 5.0	Tier 3 Review
Pacific Activewear MFG	Vietnam	Apparel & Footwear	16 days	±2.2 d	95.0%	4.7 / 5.0	Tier 1 Preferred

6. Prioritized Purchase Order Recommendations

Generated algorithmic replenishment recommendations totaling 137 purchase orders with a cumulative purchase value of $3,100,142. Sizing accounts for su

Product Name	Supplier	ROP	Inv Position	Reorder Qty	Unit Cost	Total Cost	Urgency	Est Stockout
ProBook Ultra 15" M2 Lap...	Silicon Core Sem..	466 pcs	5 pcs	820 pcs	$890.00	$729,800	Immediate	2026-10-29
AutoClean Robot Vacuum L...	Silicon Core Sem..	455 pcs	18 pcs	790 pcs	$220.00	$173,800	Immediate	2026-10-29
AirDuo Drone with 4K Gim...	Silicon Core Sem..	386 pcs	249 pcs	435 pcs	$390.00	$169,650	Upcoming (7d)	2026-11-23
Curved Ultrawide Gaming ...	Apex Tech Electr..	222 pcs	107 pcs	460 pcs	$340.00	$156,400	Upcoming (7d)	2026-11-06
BaristaTouch Espresso Ma...	SmartLiving Appl..	121 pcs	13 pcs	440 pcs	$265.00	$116,600	Immediate	2026-10-29
AeroTab 11-inch Quad HD ...	Shenzhen Micro I..	217 pcs	38 pcs	400 pcs	$260.00	$104,000	Upcoming (7d)	2026-11-03
Inflatable Stand-Up Padd...	Scandic Outdoor ..	383 pcs	75 pcs	610 pcs	$165.00	$100,650	Upcoming (7d)	2026-11-04
All-Season Ultralight 2-...	Scandic Outdoor ..	428 pcs	2 pcs	765 pcs	$125.00	$95,625	Immediate	2026-10-29
HyperDrive NVMe 2TB Exte...	Silicon Core Sem..	439 pcs	3 pcs	775 pcs	$108.00	$83,700	Immediate	2026-10-29
Down Mummy Sleeping Bag ...	Scandic Outdoor ..	429 pcs	4 pcs	765 pcs	$95.00	$72,675	Immediate	2026-10-29
Tri-Ply Stainless Steel ...	Bavaria Culinary..	183 pcs	100 pcs	435 pcs	$160.00	$69,600	Upcoming (7d)	2026-11-06
Chelsea Leather Slip-On ...	Iberian Leather ..	410 pcs	22 pcs	700 pcs	$84.00	$58,800	Immediate	2026-10-29

7. What-If Inventory Optimization Scenario Comparison

Standard baseline configuration active. Use the Interactive Simulator to project supply shocks and lead time variations.

8. Mathematical Methodology, Formulas & Operational Limitations

- Days of Inventory Cover (DOC): DOC = Available Stock / Average Daily Demand (D_avg)
- Combined Lead-Time Demand Variance: sigma_LTD = sqrt(L * sigma_D^2 + D_avg^2 * sigma_L^2)
- Dynamic Safety Stock: SS = Z_score * sigma_LTD (where Z = 1.645 at 95% cycle service level)
- Reorder Point (ROP): ROP = (D_avg * L) + SS
- Net Inventory Position (IP): IP = Current Physical Stock - Reserved Stock + Incoming Stock Orders
- Economic Reorder Sizing: ROQ = max(MOQ, ceil(Target_Stock - IP)) where Target_Stock = ROP + (Target_Cover * D_avg)

Analytical Limitations & Risk Governance Guidelines

1. Synthetic Demonstration Disclaimer: All dataset records are synthetically generated for analytical modeling and validation.

2. Historical Demand Censoring: Sales quantities reflect fulfilled demand. During historical stockouts, true unconstrained demand may be underestimated without explicit lost-demand imputation.

3. Supplier Stochasticity: Real-world lead-time variance can be non-normal due to geopolitical disruptions, port congestion, and seasonal factory closures.

4. Discretionary Approval: Large capital purchase requisitions (>$5,000) should be audited by inventory category managers prior to ERP commitment.